and a bus 1,000 miles a week, so the basic designs must be different. But on matters of reliability at least, the Defense Department seems to be much less demanding.

Cooperation in space, as I've already mentioned, is becoming an instrument of international cooperation—for example, in slowing the proliferation of strategic weapons to new nations. Rockets decommissioned because of the end of the Cold War might be gainfully employed in missions to Earth orbit, the Moon, the planets, asteroids, and comets. But all this can be accomplished without human missions to Mars.

Other justifications are offered. It is argued that the ultimate solution to world energy problems is to strip-mine the Moon, return the solar-wind-implanted helium-3 back to Earth, and use it in fusion reactors. What fusion reactors? Even if this were possible, even if it were cost-effective, it is a technology 50 or 100 years away. Our energy problems need to be solved at a less leisurely pace.

Even stranger is the argument that we have to send human beings into space in order to solve the world population crisis. But some 250,000 more people are born than die every day—which means that we would have to launch 250,000 people per day into space to maintain world population at its present levels. This appears to be beyond our present capability.

I RUN THROUGH such a list and try to add up the pros and cons, bearing in mind the other urgent claims on the federal budget. To me, the argument so far comes down to this question: Can the sum of a large number of individually inadequate justifications add up to an adequate justification?

I don't think any of the items on my list of purported justifications is demonstrably worth $500 billion or even $100 billion, certainly not in the short term. On the other hand, most of them are worth something, and if I have five items each worth $20 billion, maybe it adds up to $100 billion. If we can be clever about reducing costs